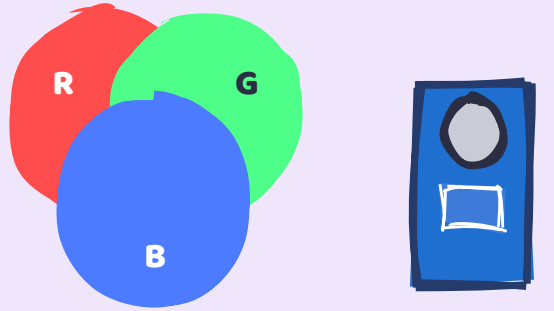
Junior Makers · Class 21

NeoPixel Light Show to the Beat

30 colour LEDs that dance to your music

RGB colours **Pixel numbers** **Sound sensor**





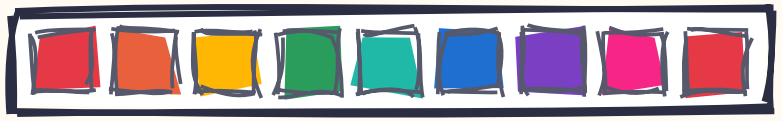
Did you know?
Big concerts use thousands of 'smart' LEDs, each with its own tiny chip. Ours work exactly the same way.

Let's think...

How do lights flash on the beat?

Wow, the lights!

At a concert or a wedding, lights change colour and flash exactly on the beat. Is someone pressing buttons super fast? Or do the lights 'hear' the music? Today **you** are the lighting designer!





A person pressing buttons?

Magic?

Code + a microphone!

Today: you program a **NeoPixel strip** to make colours, animations and a light show that reacts to **sound**.

Learning objectives

By the end of this class I can...

1	**Colour mixing:** mix any colour from **red, green and blue** numbers (0–255)	**Proof:** I made purple, orange and white on my strip
2	**Pixel numbers:** light **one LED at a time** by its number and use a loop for animations	**Proof:** my strip shows a chase effect from LED 0 to 29
3	**React to sound:** make the lights react to **claps or music** with a sound sensor	**Proof:** the strip flashes when the class claps
4	**Safe power:** explain why we limit **brightness** and use a capacitor and resistor	**Proof:** say what each protection part does

Objective 1 · Colour mixing · video

Video time!



Mixing Colors of Light

4UKIDS TV · about 3 minutes · youtu.be/QEWJMo2NH7Q

Watch for...

1. What are the three **primary colours of light**?
2. What colour do **red + green** light make?
3. What happens when all three colours shine together?

Scan to watch

After the video: Predict with your partner: what colour will red 255, green 255, blue 0 make on our strip?

Objective 1 · Colour mixing

Every colour from red, green and blue

The idea

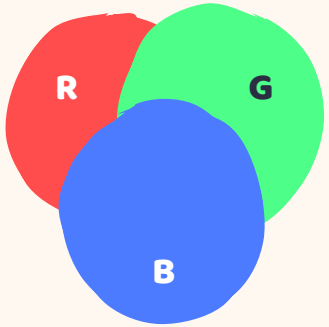
Each NeoPixel has **three tiny LEDs** inside: red, green and blue. Each can be from **0 (off) to 255 (full)**.

- red 255, green 0, blue 0 → **red**
- red 255, green 0, blue 255 → **purple**
- red 255, green 120, blue 0 → **orange**
- all 255 → **white** · all 0 → **off**

Additive

Light mixing is called ADDITIVE: adding colours makes it brighter, until all three make white.

Example: mixing paint (but with light!)



With paint, red + blue makes purple. With light, red + green makes YELLOW, which surprises everyone! Look at a TV screen with a magnifying glass: you'll see tiny red, green and blue dots.

Objective 2 · Pixel numbers

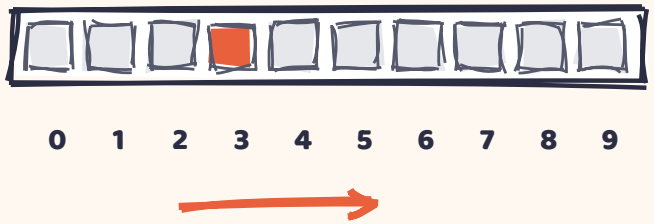
Every pixel has a number

The idea

The first LED is **pixel 0**, the last of 30 is **pixel 29**. Data travels down the strip: each LED keeps its own colour and passes the rest along.

A **for loop** counting from 0 to 29 can light them one after another: a **chase**. Turn the previous one off each time and it looks like a single light running!

Example: a stadium wave



In a stadium 'Mexican wave', person 1 stands, then person 2, then person 3... Each person only needs to know their seat number and when to stand. That's a pixel chase!

Objective 3 · React to sound

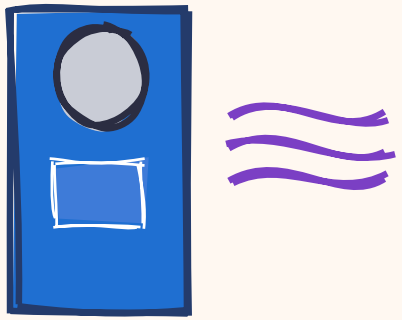
Hearing the beat

The idea

The sound sensor has a tiny **microphone** and a **knob**. When the sound is louder than the knob's setting, the **DO** pin goes HIGH and a little LED on the board lights up.

Turn the knob until the board LED is **just off** when the room is quiet. Now a clap or a drum beat flips it ON.

Example: a sensitive friend



Some friends jump at every tiny noise. Others only notice a loud bang. The knob sets how 'jumpy' the sensor is. Too jumpy = it flashes all the time. Not jumpy enough = it never hears the beat.

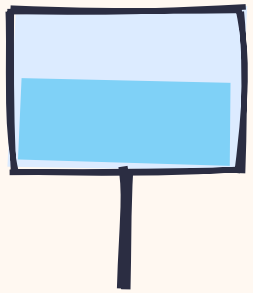
Objective 4 · Safe power

Why the extra parts?

The idea

- **Brightness 50**: each LED at full white uses about 60 mA. 30 × 60 = **1,800 mA**! USB gives about 500 mA. Too bright = the Arduino resets or gets hot.
- **100 µF capacitor**: a tiny 'tank' of energy that smooths sudden power jumps when LEDs switch on.
- **220 Ω resistor**: protects the first LED's data pin from spikes.

Example: a water tank on the roof



steady flow

Many homes in Pakistan have a roof tank, so taps keep flowing even when water pressure drops. The capacitor is a tiny tank for electricity.

Real world

Smart LEDs everywhere



Concerts
Light desks sync thousands of LEDs to the music



Buildings
Skyscrapers like the Burj Khalifa light up with LED shows



Cars
Indicator lights that 'sweep' are LED chases



Festivals
Wedding and Eid lights change colour and pattern



Where have you seen lights that change colour by themselves?

Pixels
A big LED screen is just thousands of RGB pixels arranged in a grid.

Part 2 · Make it

Now let's put it to work!

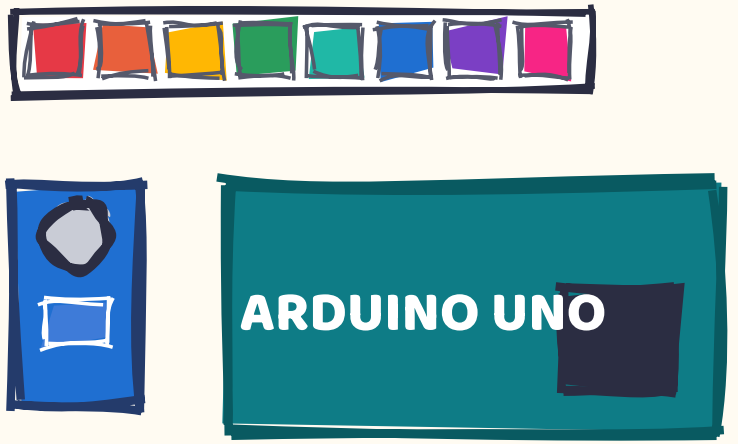
You know the ideas. Time to play, build and test. Use these slides while you work:

1. Wire the strip safely
2. Design your light show
3. Human pixels
4. Our light-show kit (per group)
5. First colours
6. Make a chase and a rainbow
7. Flash on the beat

Objective 4 · Safe power · Build step 1 of 4

Wire the strip safely

Part	Connect to
Strip 5V (red)	5V
Strip GND (white/black)	GND
Strip DIN (green)	pin 6 through 220 Ω
100 µF capacitor	across 5V and GND, long leg to 5V
Sound sensor DO / + / G	pin 2 / 5V / GND



Capacitor polarity

The stripe on the capacitor marks the – leg. Backwards, it can pop!

Objective 4 · Safe power · design

Design your light show

Part	Idea	Our plan
Intro (calm)	slow rainbow	
Beat section	flash on every clap/beat	
Finale	fast chase, then all white	
Colour theme	e.g. team colours	

Show time

Each team picks a 30-second song. Dim the lights, play the song, and run your show!

Be kind

Some people find fast flashing uncomfortable. Avoid very fast strobe effects.

Class game

Human pixels

How to play

1. Ten students stand in a line: they are pixels **0 to 9**. Each holds red, green and blue cards.
2. The 'program' (teacher) calls: **Pixel 3: red 255!** Pixel 3 holds up red.
3. Then: **All pixels: blue!** Then a chase: pixel 0, then 1, then 2…
4. Finally the class claps a beat: on every clap, all pixels flash a colour.

Then discuss

Why did we number the pixels from 0, not 1? What instruction made the 'chase'?



Big idea

Every pixel has a number. Loops let us change them one after another, very fast.

Build it

Our light-show kit (per group)

Arduino Uno
The light-show brain

NeoPixel strip (30)
WS2812B smart LEDs

KY-037 sound sensor
Hears claps and beats

100 µF capacitor
Smooths the power

220 Ω resistor
Protects the data line

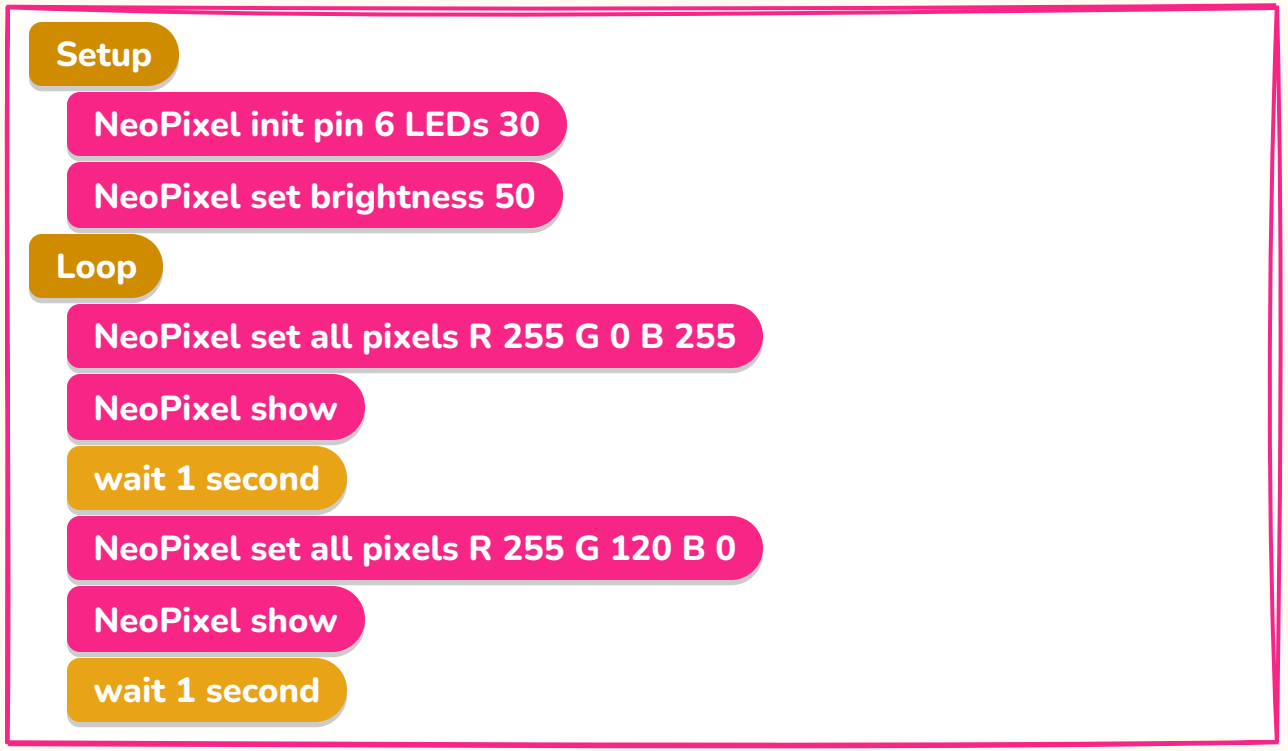
Laptop
ArduinoBlocks

Also: breadboard and jumper wires. The strip has an arrow: data flows **in** the direction it points.

Too bright!
30 LEDs at full white can pull nearly 2 A, more than USB can give. We keep brightness at 50 out of 255.

Objective 1 · Colour mixing · Build step 2 of 4

First colours



Colour	R	G	B
purple	255	0	255
orange	255	120	0
cyan			
yellow			
my colour: ____			

Don't forget show!
Setting a colour only changes memory. 'show' sends it to the LEDs.

Objective 2 · Pixel numbers · Build step 3 of 4

Make a chase and a rainbow

```
Loop
  count with i from 0 to 29
    NeoPixel set pixel i R 0 G 0 B 255
    NeoPixel show
    wait 30 milliseconds
    NeoPixel set pixel i R 0 G 0 B 0
(then try: rainbow)
  count with i from 0 to 29: set pixel i to hue i × 8
```

Speed

Change 30 ms to 10 ms for a fast zoom, or 100 ms for a slow walk.

Bounce

Add a second loop from 29 back down to 0. Now it bounces like a scanner!

MAKE

Objective 3 · React to sound · Build step 4 of 4

Flash on the beat

```
Loop
  if digital read pin 2 = HIGH then
    NeoPixel set all pixels random colour
    NeoPixel show
    wait 80 milliseconds
  else
    NeoPixel set all pixels R 0 G 0 B 0
    NeoPixel show
```

Test

Clap! Then play music with a strong drum beat from a phone, close to the sensor.

Flashing all the time?
Turn the sensor's knob a little until its small LED goes off in silence.

Test like an engineer

Bug hunt

Bug	Likely cause	Did it happen?
Nothing lights	data wire on wrong end (arrow!) or wrong pin	
Only the first LED works	LED count set to 1 in init	
Colours are wrong (red shows green)	strip type is GRB: change setting	
Arduino keeps resetting	brightness too high	
Flashes with no sound	sensor knob too sensitive	

Debug checklist

Test with brightness 20 and one colour first. Add effects one at a time.

If the strip flickers randomly, check the GND wire and the 220 Ω resistor.

Level up!

Light designer challenges

Choose your challenge



VU meter: use the sensor's analog pin A0: louder = more LEDs light up.



Sparkle: light random pixels in random colours, then fade them.



Two chases: two lights running in opposite directions.



Heartbeat: make the whole strip 'breathe' by changing brightness slowly.

Think like an engineer

Why does red + green light make yellow, but red + green paint makes brown?

Boss level

Make a strip that shows a different effect each time you clap (count the claps).

Objective check

Can I really do it? Prove it.

01	Tell us the R, G, B numbers for purple and for white.	○ ○ ○
02	Show your chase. Which block makes it go from pixel 0 to 29?	○ ○ ○
03	Clap! Show your lights reacting. How did you set the sensor knob?	○ ○ ○
04	What do the capacitor, the resistor and brightness 50 each protect?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Light-show festival: dim the room, each team runs a 30-second show.

Show what you know

What did we learn?

- [x] Every colour = red + green + blue (0–255 each)
- [x] Pixels are numbered from 0; loops make animations
- [x] A sound sensor's knob sets how loud a beat must be
- [x] Keep brightness low and use a capacitor and resistor
- [x] Always call 'show' to send colours to the strip

Exit ticket: what colour is R 0, G 255, B 255?

At home

Look at a phone or TV screen with a magnifying glass. Draw the tiny coloured dots you see.



Next class: 8×8 LED Matrix Emoji Display!